

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1408

Lutesville 138 kV
March 05, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1408. The Interconnection Customer proposes a 48.8 MW battery energy storage facility interconnecting at the Lutesville 138 kV substation in the Pennsylvania area. The request was received on February 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 527 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,365,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: W Campus 345 kV/138 kV Autotransformer
Contingency: Edes Tap 2 / W Campus 345 KV
Loading: 106.8% (Rating: 559 MVA)
DFAX: 13.9%
Pre-project loading: 105.6%
2. Facility: W Campus / Big Hollow Rd 138 KV line
Contingency: W Campus / Edes Tap 2 345 KV
Loading: 105.9% (Rating: 588 MVA)
DFAX: 13.9%
Pre-project loading: 104.8%
3. Facility: Little Chute / Talenenergy Martins Crk Llc arrisburg Substation 345 KV
Contingency: Fort Humbug / Coronet 500 KV
Loading: 102.2% (Rating: 2,249 MVA)
DFAX: 9.8%
Pre-project loading: 102.0%
4. Facility: Hastings / Elk City Llc 345 KV Line
Contingency: Little Chute / Fort Humbug Substation 345 KV
Loading: 101.9% (Rating: 1,505 MVA)
DFAX: 5.2%
Pre-project loading: 101.7%
5. Facility: Seward Substation / Kearny 345 KV
Contingency: Armstrong / Ft Gibson 500 KV Line
Loading: 100.3% (Rating: 1,386 MVA)
DFAX: 12.3%
Pre-project loading: 99.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
West Campus 345/138kV Transformer	Replace transformer with higher-rated unit	\$20,921,000	10% (\$2,129,000)
Big Holow Rd / West Campus 138 KV	Reconductor line	\$696,000	10% (\$71,000)
Little Chute / Talenenergy Martins Crk Llc arisburg Substation 345 KV	Reconductor line	\$5,317,000	1% (\$33,000)
Hastings / Elk City Llc 345 KV Line	Reconductor line	\$10,447,000	1% (\$79,000)
Seward / Kearny 345 KV line	Reconductor line	\$4,197,000	1% (\$53,000)
	Total allocated to project		\$2,365,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.